



FORM VI

MATHEMATICS

Examination date

Tuesday 1st August 2006

Time allowed

Three hours (plus 5 minutes reading time)

Instructions

- All ten questions may be attempted.
- All ten questions are of equal value.
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Collection

- Write your candidate number clearly on each booklet.
- Hand in the ten questions in a single well-ordered pile.
- Hand in a booklet for each question, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Keep the printed examination paper and bring it to your next Mathematics lesson.

Checklist

- SGS booklets: 10 per boy. A total of 1250 booklets should be sufficient.
- Candidature: 105 boys.

Examiner

REN

QUESTION ONE (12 marks) Use a separate writing booklet.

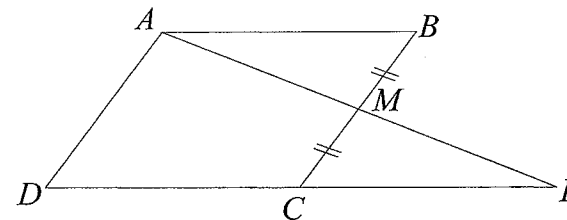
Marks

- (a) Solve $x^2 - 3x - 10 = 0$. 1
- (b) Convert $\frac{5\pi}{9}$ radians to degrees. 1
- (c) Find a primitive of $x^5 - 1$. 1
- (d) Use your calculator to find $\frac{15.7}{\sqrt{1.6 + 2.9}}$, giving your answer correct to one decimal place. 2
- (e) Write $\frac{7}{\sqrt{5} - 2}$ with a rational denominator. 2
- (f) Solve $|x - 3| = 1$. 1
- (g) Sketch the graph of $x^2 + y^2 = 9$, showing all x and y -intercepts. 2
- (h) A man's weekly income is \$1650. If he receives a 4% wage increase, find his new weekly income. 2

QUESTION TWO (12 marks) Use a separate writing booklet.

Marks

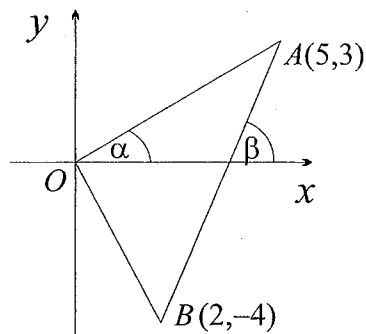
(a)



The diagram above shows the parallelogram $ABCD$ with M the midpoint of BC . The intervals AM and DC are produced to meet at P .

- (i) Prove that $\triangle ABM \equiv \triangle PCM$. 2
- (ii) Hence prove that $ABPC$ is a parallelogram. 2

(b)



The diagram above shows $\triangle AOB$ with A and B the points $(5,3)$ and $(2,-4)$ respectively. The angle of inclination of OA is α and the angle of inclination of AB is β .

- (i) Write down the gradients of OA and AB . 2
- (ii) Hence find α and β , both correct to the nearest degree. 2
- (iii) Find the length of OA . 1
- (iv) Find the length of AB . 1
- (v) Find the area of $\triangle AOB$. Give your answer correct to two significant figures. 2

QUESTION THREE (12 marks) Use a separate writing booklet.

Marks

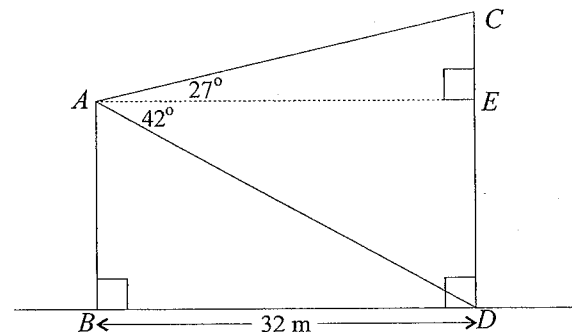
- (a) Differentiate the following with respect to x :
 - (i) $\log_e(4x+3)$ 1
 - (ii) $x \sin x$ 1
- (b) (i) Find $\int e^{6x} dx$. 1
- (ii) Evaluate $\int_1^9 \sqrt{x} dx$. 2
- (c) The equation of a parabola is given by $(x-1)^2 = 8y$.
 - (i) Write down the coordinates of the vertex. 1
 - (ii) Write down the focal length. 1
 - (iii) Sketch the graph of the parabola clearly showing the focus and directrix. 3
- (d) Solve the equation $\tan x = \sqrt{3}$, for $0 \leq x \leq 2\pi$. 2

Exam continues overleaf ...

QUESTION FOUR (12 marks) Use a separate writing booklet.

Marks

- (a) Find the equation of the tangent to the curve $y = x^2 - 3x$ at the point $(1, -2)$. 3
- (b)



In the diagram above, two vertical buildings AB and CD are 32 metres apart. From A , the angle of elevation of C is 27° and the angle of depression to D is 42° .

- (i) Find the height of building AB , correct to two significant figures. 2
- (ii) Find the height of building CD , correct to two significant figures. 1
- (c) Find all solutions of the equation $x^4 - 7x^2 + 12 = 0$. 3
- (d) The table below shows the values of the function $f(x)$ for five values of x : 3

x	4	4.5	5	5.5	6
$f(x)$	1.3	2.9	0.7	-0.2	-1.1

Use Simpson's rule with these five function values to find an estimate for $\int_4^6 f(x) dx$.
Give your answer correct to one decimal place.

QUESTION FIVE (12 marks) Use a separate writing booklet.

Marks

- (a) Write down the domain and range of the function $y = 4 \sin x$. 2
- (b) Consider the quadratic equation $x^2 - kx + (k+3) = 0$.
 - (i) Find the discriminant and write it in simplest form. 1
 - (ii) For what values of k does the equation have no real roots? 2
 - (iii) If the product of the roots is equal to three times the sum of the roots, find the value of k . 2

Exam continues next page ...

- (c) In order to study the history of the Earth's climate, a team of scientists drilled an "ice core" in the Antarctic ice sheet. They drilled 5 metres on the first day, a further 7 metres on the second day, a further 9 metres on the third day and so on.

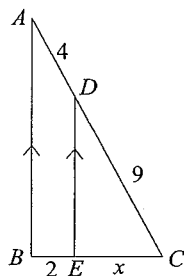
- (i) Find how many metres they drilled on the 40th day. 1
 (ii) Find how deep they had drilled after 40 days. 1
 (iii) Find how many days it took to drill to a depth of 480 metres. 3

QUESTION SIX (12 marks) Use a separate writing booklet.

Marks

- (a) The equation of a curve is given by $y = e^x - x$. Find the stationary point on the curve and determine its nature. 3
 (b) On a certain island, the population P of rabbits is increasing so that after time t weeks the value of P is given by $P = Ae^{kt}$, where A and k are constants. When the population was first measured there were 200 rabbits on the island, and 5 weeks later there were 750 rabbits.
 (i) Write down the value of A . 1
 (ii) Find the exact value of k . 1
 (iii) Find the number of rabbits on the island after 12 weeks. (Give your answer correct to three significant figures.) 1
 (iv) Find how long it will take for the rabbit population to reach 100 000. (Give your answer correct to the nearest week.) 2
 (v) Find the rate at which the rabbit population is increasing after 4 weeks. (Give your answer correct to the nearest whole number.) 2

(c)



The diagram above shows $\triangle ABC$ with points D and E on AC and BC respectively so that $DE \parallel AB$. $AD = 4$, $DC = 9$, $BE = 2$ and $EC = x$. Find the value of x , giving reasons.

Exam continues overleaf ...

QUESTION SEVEN (12 marks) Use a separate writing booklet.

Marks

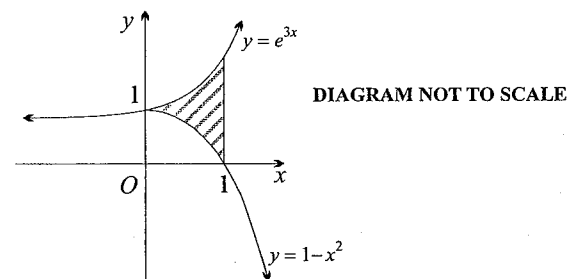
- (a) A particle P is moving in a straight line so that its velocity v metres per second after t seconds is given by $v = 12 - 4t$. Initially P is 3 metres to the right of the origin O .
 (i) Find the initial velocity and acceleration of P . 2
 (ii) If the displacement of P from O is x metres, find an expression for x in terms of t . 2
 (iii) Find when and where P is stationary. 2
 (iv) Sketch the graph of $v = 12 - 4t$, for $0 \leq t \leq 5$. 1
 (v) Hence, or otherwise, find the total distance travelled by P during the first five seconds. 2

- (b) Solve the equation $\log_2 x + 1 = \log_2 \sqrt{x}$. 3

QUESTION EIGHT (12 marks) Use a separate writing booklet.

Marks

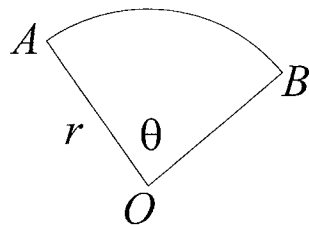
- (a) Differentiate $\ln(\cos x)$ and express your answer in simplest form. 2
 (b) 3



The diagram above shows the region enclosed between the two curves $y = e^{3x}$ and $y = 1 - x^2$ and the line $x = 1$. Find the area of this region.

Exam continues next page ...

(c)



The diagram above shows a sector AOB , with radius r cm and with $\angle AOB = \theta$ radians. The area of the sector AOB is 40 cm^2 .

- (i) If P cm is the perimeter of sector AOB , show that $P = 2r + \frac{80}{r}$. 1
- (ii) Find the value of r and of θ for which P is a minimum. Justify your answer. 3
- (iii) Suppose now that, instead of 40 cm^2 , the area of sector AOB is $x \text{ cm}^2$.
Find the value of θ for which P is a minimum. 2
- (iv) What general conclusion can you draw from part (iii)? 1

QUESTION NINE (12 marks) Use a separate writing booklet.

Marks

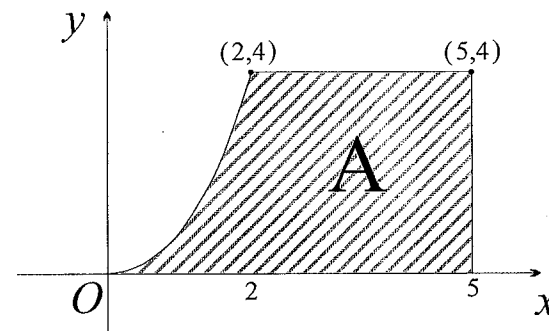
- (a) (i) Sketch the graph of $y = \cos 2x$, for $0 \leq x \leq 2\pi$. (Your diagram should take up at least a quarter of a page.) 2
- (ii) Copy and complete the following table of values for $y = \frac{1}{4}x$. (Give the values correct to one decimal place where necessary.) 1

x	0	π	2π
y			

- (iii) Use the table to sketch the graph of $y = \frac{1}{4}x$ on the same diagram as part(i). 1
- (iv) Hence determine the number of solutions of the equation $4 \cos 2x - x = 0$.
(Note: You do not have to solve this equation.) 2

Exam continues overleaf ...

(b)



A function $f(x)$ is defined as follows:

$$f(x) = \begin{cases} x^2, & \text{for } 0 \leq x \leq 2, \\ 4, & \text{for } 2 < x \leq 5. \end{cases}$$

In the diagram above the shaded region A is bounded by the graph of $f(x)$, the x -axis and the line $x = 5$.

- (i) Find the volume of the solid formed when the shaded region A is rotated about the x -axis. 3
- (ii) If the shaded region A is now rotated about the y -axis, find the volume of the solid formed. 3

Exam continues next page ...

QUESTION TEN (12 marks) Use a separate writing booklet.

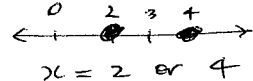
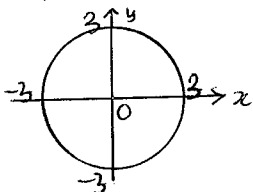
Marks

- (a) (i) Emily borrows \$80 000 in order to buy a home unit. The interest rate is 6% per annum reducible and the loan is to be repaid in equal monthly repayments over 20 years with the interest calculated monthly.

Let $\$A_n$ be the amount owing after the n th repayment.

- (α) Write down expressions for $\$A_1$ and $\$A_2$, the amounts owing after the first and second repayments respectively. 1
- (β) Show that the amount of each monthly repayment is \$573.14 (correct to the nearest cent). 2
- (ii) After $2\frac{1}{2}$ years (i.e. 30 repayments) the interest rate rises to $7\frac{1}{2}\%$. Find the new monthly repayment, correct to the nearest cent. (Assume that the period of the loan is still 20 years.) 3
- (b) (i) Show that $\frac{d}{dx}(\sec x) = \sec x \tan x$. 2
- (ii) Show that $\frac{d}{dx}(\sec x + \tan x) = \sec x(\sec x + \tan x)$. 1
- (iii) Hence, or otherwise, evaluate $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \sec x \, dx$. 3

END OF EXAMINATION**Q1**

- (a) $(x-5)(x+2) = 0$
 $x = 5 \text{ or } -2$ ✓
- (b) $\frac{5\pi}{9} \times \frac{180}{\pi} = 100^\circ$ ✓
- (c) $\int (x^5 - 1) \, dx = \frac{x^6}{6} - x + c$ ✓
- (d) 7.4 (1 dec. place) ✓✓
- (e) $\frac{7}{\sqrt{5}-2} \times \frac{\sqrt{5}+2}{\sqrt{5}+2} = \frac{7(\sqrt{5}+2)}{5-4}$
 $= 7(\sqrt{5}+2)$ ✓
- (f)  $x = 2 \text{ or } 4$ ✓
- (g)  ✓✓
- (h) New income = 1.04×1650
 $= \$1716$ ✓

Q2

- (a) (i) In $\triangle ABM$, $\triangle PCM$
1. $BM = MC$ (given) ✓
2. $\angle AMB = \angle PMC$ (vert. opp. $\angle$ s) ✓
3. $\angle BAM = \angle MCP$ (alt $\angle$ s $AB \parallel CP$) ✓
 $\triangle ABM \equiv \triangle PCM$ (ASA) ✓
- (ii) $AM = MP$ (matching sides) ✓
 $ABPC$ is a parallelogram (diag. BC, AP bisect each other at M) ✓

(b) (i) $\text{grad } OA = \frac{3}{5}$ ✓
 $\text{grad } AB = \frac{3-(-4)}{5-2}$
 $= \frac{7}{3}$ ✓

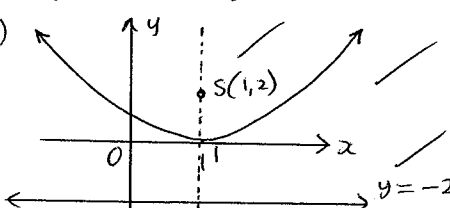
(ii) $\tan \alpha = \frac{3}{5}$ ✓
 $\alpha = 31^\circ$ ✓
 $\tan \beta = \frac{7}{3}$ ✓
 $\beta = 67^\circ$ ✓

(iii) $OA = \sqrt{5^2 + 3^2}$
 $= \sqrt{34}$ ✓

(iv) $AB = \sqrt{(5-2)^2 + (3-(-4))^2}$
 $= \sqrt{3^2 + 7^2}$
 $= \sqrt{58}$ ✓

(v) $\text{Area } \triangle AOB = \frac{1}{2} \times \sqrt{58} \times \sqrt{34} \times \sin 36^\circ$
 $= 13 \text{ units}^2$ ✓

Q3

- (a) (i) $\frac{4}{4x+3}$ ✓
 (ii) $x \times \cos x + 1 \times \sin x$
 $= x \cos x + \sin x$ ✓
- (b) (i) $\frac{1}{6} e^{bx} + c$ ✓
 (ii) $\frac{2}{3} \left[x^{\frac{3}{2}} \right]^9$ ✓
 $= \frac{2}{3} (9^{\frac{3}{2}} - 1^{\frac{3}{2}})$ ✓
 $= \frac{52}{3}$ ✓
- (c) (i) Vertex $(1, 0)$ ✓
 (ii) Focal length = 2 ✓
- (iii)  ✓
- (d) related angle = $\frac{\pi}{3}$ ✓
 $x = \frac{\pi}{3}$ or $\frac{4\pi}{3}$ ✓

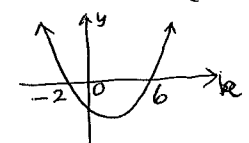
Q4

- (a) $\frac{dy}{dx} = 2x - 3$ ✓
 At $(1, -2)$, grad = $2 \times 1 - 3 = -1$ ✓
 Eqn of tangent: $y + 2 = -1(x - 1)$ ✓
 $y = -x - 1$ ✓
- (b) (i) $\tan 42^\circ = \frac{AB}{32}$ ✓
 $AB = 32 \times \tan 42^\circ$ ✓
 $\approx 29 \text{ m}$ ✓
- (ii) $\tan 27^\circ = \frac{CE}{32}$ ✓
 $CE = 32 \times \tan 27^\circ$ ✓
 $CD = CE + AB$ ✓
 $\approx 45 \text{ m}$ ✓

- (c) let $a = x^2$
 $a^2 - 7a + 12 = 0$ ✓
 $(a - 4)(a - 3) = 0$ ✓
 $a = 4$ or 3 ✓
 $x^2 = 4$ or $x^2 = 3$ ✓
 $x = 2, -2, \sqrt{3}$ or $-\sqrt{3}$ ✓

(d) $\int_4^6 f(x) dx = \int_4^5 f(x) dx + \int_5^6 f(x) dx$ ✓
 $\approx \frac{1}{6} [f(4) + 4 \times f(4.5) + f(5)] + \frac{1}{6} [f(5) + 4 \times f(5.5) + f(6)]$ ✓
 $= \frac{1}{6} [1.3 + 4 \times 2.9 + 2 \times 0.7 + 4 \times (-0.2) - 1.1]$ ✓
 ≈ 2.1 ✓

Q5

- (a) Domain: $x \in \mathbb{R}$ ✓
 Range: $-4 \leq y \leq 4$ ✓
- (b) (i) $\Delta = (-4)^2 - 4 \times 1 \times (k+3)$ ✓
 $= k^2 - 4k - 12$ ✓
- (ii) $k^2 - 4k - 12 < 0$ ✓
 $(k-6)(k+2) < 0$ ✓
 $-2 < k < 6$ ✓
 ✓
- (iii) $3k = k + 3$ ✓
 $2k = 3$ ✓
 $k = 1\frac{1}{2}$ ✓
- (c) (i) $5 + 7 + 9 + 11 + \dots$ ✓
 (AP, $a = 5, d = 2$) ✓
 $T_{40} = 5 + 39 \times 2$ ✓
 $= 83 \text{ metres}$ ✓
- (ii) $S_{40} = \frac{40}{2} (5 + 83)$ ✓
 $= 1760 \text{ metres}$ ✓
- (iii) $480 = \frac{n}{2} [10 + 2(n-1)]$ ✓
 $= n(5 + n - 1)$ ✓
 $n^2 + 4n - 480 = 0$ ✓
 $(n + 24)(n - 20) = 0$ ✓
 $n = 20$ ✓
 i.e. 20 days. ✓

Q6

- (a) $y' = e^x - 1$ ✓
 At start pt. $e^x - 1 = 0$ ✓
 $e^x = 1$ ✓
 $x = 0$ ✓
 when $x = 0$, $y = e^0 - 0 = 1$ ✓
 $y' = 0$ at $(0, 1)$ ✓

$y'' = e^x$

- At $(0, 1)$, $y'' = e^0 = 1$ ✓
 since $y'' > 0$, $(0, 1)$ is a rel. min. ✓
- (b) (i) $A = 200$ ✓
 (ii) $750 = 200 e^{5k}$ ✓
 $k = \frac{1}{5} \ln \frac{15}{4}$ ✓
- (ii) when $t = 12$, $P = 200 e^{\frac{12}{5} \ln \frac{15}{4}}$ ✓
 $\approx 4770 \text{ rabbits}$ ✓
- (iv) $100000 = 200 e^{kt}$ ✓
 $t = \frac{\ln 500}{k}$ ✓
 $\approx 24 \text{ weeks}$ ✓
- (v) $\frac{dp}{dt} = A k e^{kt}$ ✓
 when $t = 4$, $\frac{dp}{dt} = 200 k e^{4k}$ ✓
 $\approx 152 \text{ rabbits/week}$ ✓
- (c) $\frac{EC}{EB} = \frac{CD}{DA}$ (Interval // to one side of triangle divides other two sides in proportion) ✓
 $\frac{x}{2} = \frac{9}{4}$ ✓
 $x = 4\frac{1}{2}$ ✓

Q7

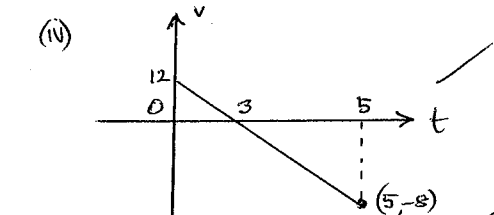
(a) (i) When $t=0$, $v = 12 - 4 \times 0$
 $= 12 \text{ m/s}$
 $\frac{dv}{dt} = -4 \text{ m/s}$

(ii) $x = 12t - 2t^2 + c$
 $t=0$, $x=3$, $3 = 0 - 0 + c$
 $c = 3$

$x = 12t - 2t^2 + 3$

(iii) Stationary when $12 - 4t = 0$
 $t = 3 \text{ seconds}$

When $t=3$, $x = 12 \times 3 - 2 \times 3^2 + 3$
 $= 21$
 Stationary when $x = 21 \text{ m}$



(v) Distance $= \frac{12 \times 3}{2} + \frac{2 \times 8}{2}$
 $= 26 \text{ m}$

(b) $\log_2 x + 1 = \log_2 \sqrt{x}$
 $\log_2 x + 1 = \frac{1}{2} \log_2 x$
 $\frac{1}{2} \log_2 x = -1$
 $\log_2 x = -2$
 $x = 2^{-2}$
 $x = \frac{1}{4}$

Q8

(a) $\frac{d}{dx} (\ln \cos x) = \frac{1}{\cos x} \times (-\sin x)$
 $= -\tan x$

(b) Area $= \int_0^1 (e^{3x} - 1 + x^2) dx$
 $= \left[\frac{1}{3} e^{3x} - x + \frac{x^3}{3} \right]_0^1$
 $= \frac{1}{3} e^3 - 1 + \frac{1}{3} - \frac{1}{3} e^0$
 $= \left(\frac{1}{3} e^3 - 1 \right) \text{ units}^2$

(c) (i) $p = 2r + r\theta$
 Now, $\frac{1}{2} r^2 \theta = 40$
 $\theta = \frac{80}{r^2}$

$p = 2r + \frac{80}{r^2} \times r$
 $= 2r + \frac{80}{r}$

(ii) $\frac{dp}{dr} = 2 - \frac{80}{r^2}$
 At min, $2 - \frac{80}{r^2} = 0$
 $2r^2 = 80$
 $r^2 = 40$
 $r = 2\sqrt{10}$, $\theta = \frac{80}{(2\sqrt{10})^2} = 2$

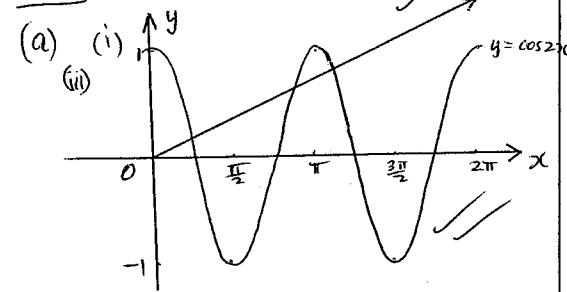
$\frac{d^2p}{dr^2} = \frac{160}{r^3}$
 when $r = 2\sqrt{10}$, $\frac{d^2p}{dr^2} = \frac{160}{(2\sqrt{10})^3} > 0$
 There is a min value

(iii) $\frac{1}{2} r^2 \theta = x$
 $\theta = \frac{2x}{r^2}$
 $p = 2r + \frac{2x}{r^2} \times r$
 $= 2r + \frac{2x}{r}$

$\frac{dp}{dr} = 2 - \frac{2x}{r^2}$
 At min, $2 - \frac{2x}{r^2} = 0$
 $2r^2 = 2x$
 $r = \sqrt{x}$
 when $r = \sqrt{x}$, $\theta = \frac{2x}{(\sqrt{x})^2} = 2$

(iv) The perimeter of the sector has a minimum value when $\theta = 2$ no matter what the area.

Q9



(ii)

x	0	π	2π
y	0	0.8	1.6

(iv) $4 \cos 2x - x = 0$
 $\cos 2x = \frac{x}{4}$
 No of solutions = 3
 (For $0 \leq x \leq 2\pi$)

(b) (i) Volume $= \int_2^5 4^2 x \pi dx + \int_0^2 \pi (x^2)^2 dx$

$= \pi \left[16x \right]_2^5 + \frac{\pi}{5} \left[x^5 \right]_0^2$
 $= \pi (80 - 32) + \frac{\pi}{5} \times 32$
 $= 48\pi + \frac{32\pi}{5}$
 $= \frac{272\pi}{5} \text{ units}^3$

(ii) Volume $= \int_0^4 \pi (5^2) dy - \int_0^4 \pi y dy$
 $= \pi \int_0^4 (25 - y) dy$
 $= \pi \left[25y - \frac{y^2}{2} \right]_0^4$
 $= \pi \left[100 - \frac{16}{2} - 0 \right]$
 $= 92\pi \text{ units}^3$

Q10

(a) (i) let \$M be monthly repayment

$$A_1 = 80000 \times 1.005 - M$$

$$A_2 = 80000 \times 1.005^2 - M \times 1.005 - M$$

$$(b) A_{240} = 80000 \times 1.005^{240} - M \times 1.005^{239}$$

$$- M$$

$$= 0 \text{ after 20 years}$$

$$M(1 + 1.005 + \dots + 1.005^{239}) = 80000 \times 1.005^{240}$$

$$80000 \times 1.005^{240}$$

$$M = \frac{80000 \times 1.005^{240}}{\left(\frac{1.005^{240} - 1}{1.005 - 1} \right)}$$

$$\approx \$573.14$$

(ii) Let new monthly repayment be \$P

Amount owing after 30 repayments:

$$A_{30} = 80000 \times 1.005^{30} - 573.14 \times 1.005^{29}$$

$$- \dots - 573.14$$

$$= 80000 \times 1.005^{30} - (573.14 + 573.14 \times 1.005 + \dots + 573.14 \times 1.005^{29})$$

$$= 80000 \times 1.005^{30} - \left[\frac{573.14(1.005^{30} - 1)}{1.005 - 1} \right]$$

$$\approx \$74411.04$$

$$A_{210} = 74411.04 \times 1.00625^{210} - P \times 1.00625^{209}$$

$$- P \times 1.00625^{208} - \dots - P$$

$$\text{After 20 years } A_{210} = 0$$

$$74411.04 \times 1.00625^{210}$$

$$= P(1 + 1.00625 + 1.00625^2 + \dots + 1.00625^{209})$$

$$P = \frac{74411.04 \times 1.00625^{210}}{\left(\frac{1.00625^{210} - 1}{1.00625 - 1} \right)}$$

$$\approx \$637.30$$

(b) (i) let $y = \sec x = (\cos x)^{-1}$

$$\frac{dy}{dx} = -(\cos x)^{-2} \times (-\sin x)$$

$$= \frac{1}{\cos^2 x} \times \sin x$$

$$= \frac{1}{\cos x} \times \frac{\sin x}{\cos x}$$

$$= \sec x \tan x$$

(ii) let $y = \sec x + \tan x$

$$\frac{dy}{dx} = \sec x \tan x + \sec^2 x$$

$$= \sec x (\tan x + \sec x)$$

(iii) $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \sec x \, dx$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sec x (\sec x + \tan x)}{\sec x + \tan x} \, dx$$

$$= \left[\ln(\sec x + \tan x) \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}}$$

$$= \ln(\sec \frac{\pi}{3} + \tan \frac{\pi}{3}) - \ln(\sec \frac{\pi}{4} + \tan \frac{\pi}{4})$$

$$= \ln(2 + \sqrt{3}) - \ln(\sqrt{2} + 1)$$

$$= \ln\left(\frac{2 + \sqrt{3}}{\sqrt{2} + 1}\right)$$

[Must be in this form]